

Prediction of CYP450 Metabolizer Phenotypes from SNP Array Data Using GenoStaR: A Pharmacogenetic study in the MadManic cohort

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Introduction

Pharmacogenetics plays a key role in personalized medicine by linking genetic variation to differences in drug response. Variants in cytochrome P450 genes influence both drug efficacy and adverse effects. Since CYP450 enzymes metabolize 70-80% of clinically psychotropic drugs, accurate genotype-to-phenotype translation is essential for optimizing pharmacological therapies. In this context, GenoStaR is an R package that converts genotypes into star alleles and predicts metabolic phenotypes for major CYP450 genes using SNPs, insertion/deletion variants, and structural variants.

Objective

Validate GenoStaR in 660 genotyped individuals from the MadManic cohort to obtain diplotypes, activity scores, and metabolic phenotypes for the main CYP450 genes.

Methods

The methods from this study are summarized in Figure 1.

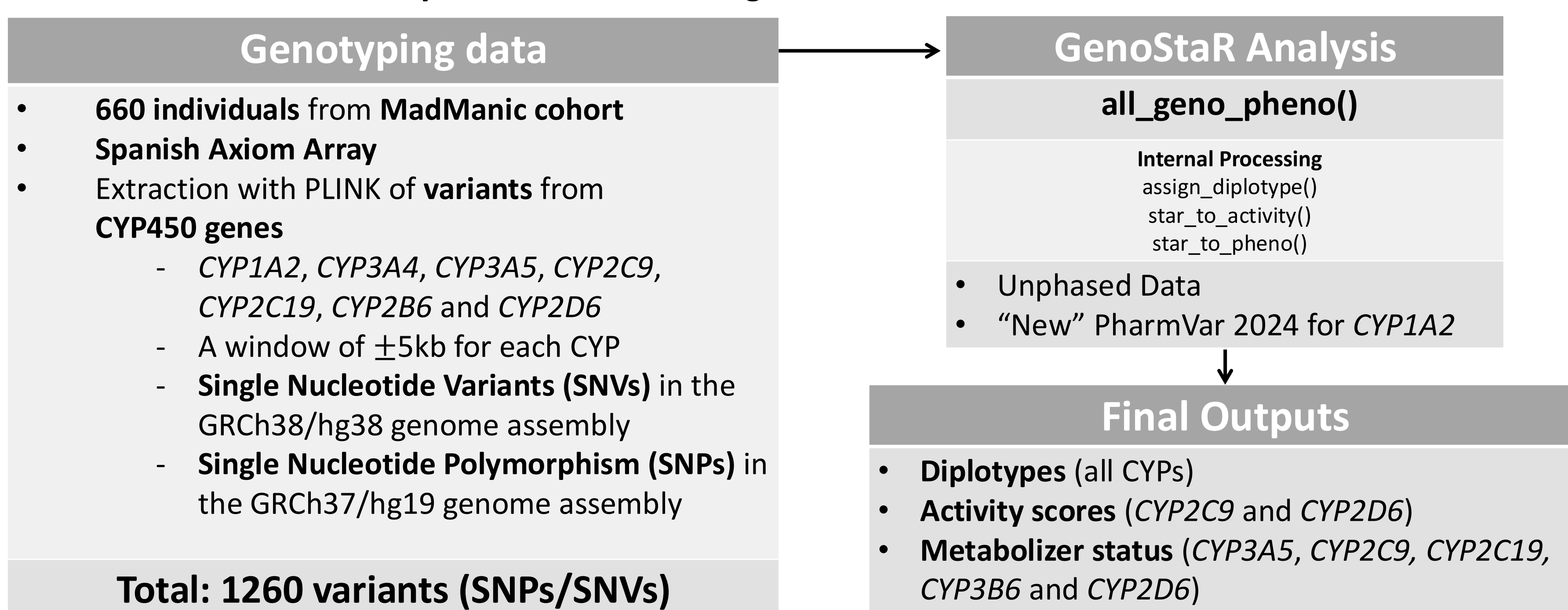


Figure 1. Workflow. Diagram summarizes the analysis pipeline applied to 660 individuals from MadManic cohort genotyped with the Spanish Axiom Array. Variants from seven CYP450 genes were extracted using PLINK and processed through the GenoStaR function all_genos_pheno(). Final outputs include diplotypes, activity scores, and metabolizer status.

Results: validation and performance of GenoStaR

The variants not genotyped in our cohort but required by GenoStaR for the European population was low (Table 1), suggesting adequate coverage of genotype data from the Spanish Axiom array and low error for most of the CYPs.

Call rates above 95% were obtained for SNPs of *CYP1A2*, *CYP2B6*, *CYP2C9*, *CYP3A4*, *CYP2C19* and *CYP3A5* (Table 1), demonstrating a high inference estimations for diplotypes. Moreover, comparison of star allele frequencies (Figure 2), activity scores, and metabolic phenotypes (Figure 3) with ClinPGx European population frequencies showed high concordance with results from our cohort.

In contrast, *CYP2D6* showed lower call rate (Table 1) and greater discrepancies with ClinPGx (Table 1, Figure 2 and Figure 3), likely due to limitations in the assignment of complex diplotypes in this gene.

Gene	Total SNPs/SNVs	Common SNPs/SNVs	Axiom Array SNPs/SNVs	GenoStaR SNPs/SNVs	GenoStaR error (%)	GenoStaR call rate (%)	ClinPGx concordance
<i>CYP1A2</i>	106	22	60	24	0.19%	100%	Very high
<i>CYP3A4</i>	112	18	70	24	0.55%	99%	Very high
<i>CYP3A5</i>	78	4	73	1	0%	100%	Very high
<i>CYP2C19</i>	338	24	303	11	0.03%	98%	High
<i>CYP2C9</i>	342	41	262	39	0.03%	99%	High
<i>CYP2B6</i>	280	30	232	18	20.05%	96%	Very high
<i>CYP2D6</i>	227	40	77	110	30.65%	48%	Medium

Table 1. Summary of the CYP genes analyzed. Including variants used by GenoStaR, those genotyped in our study, and the overlap between them; estimated error from missing variants; diplotype call rate of GenoStaR; concordance with ClinPGx population frequency data.

Star alleles frequencies

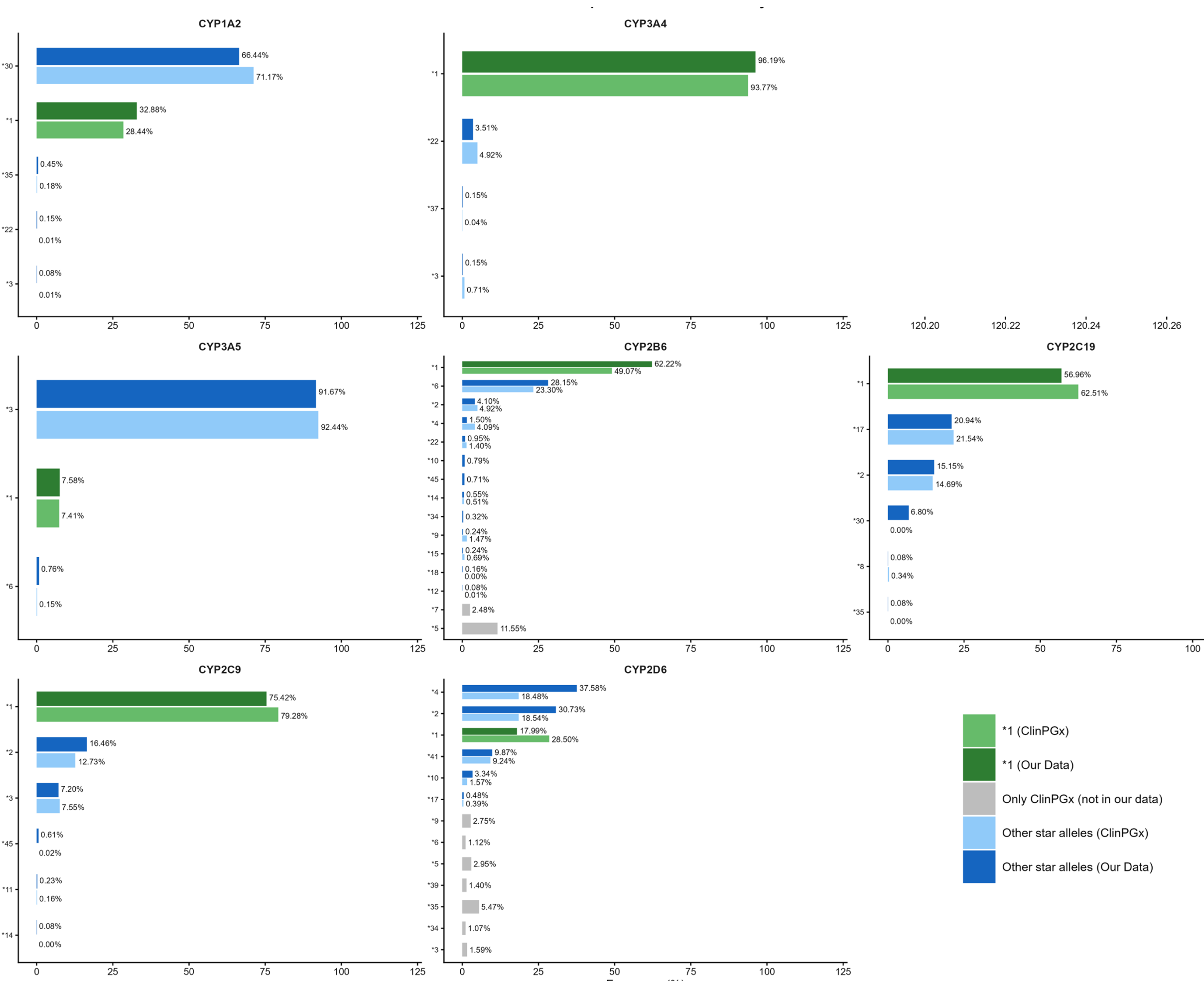


Figure 2. Comparison of CYP enzyme star alleles frequencies between our cohort (derived using the GenoStaR R package) and ClinPGx population data. Bars show relative frequencies (%) of *1 and other star alleles for each CYP enzyme; lighter shades represent our cohort and darker shades ClinPGx.

Metabolizer phenotype frequencies

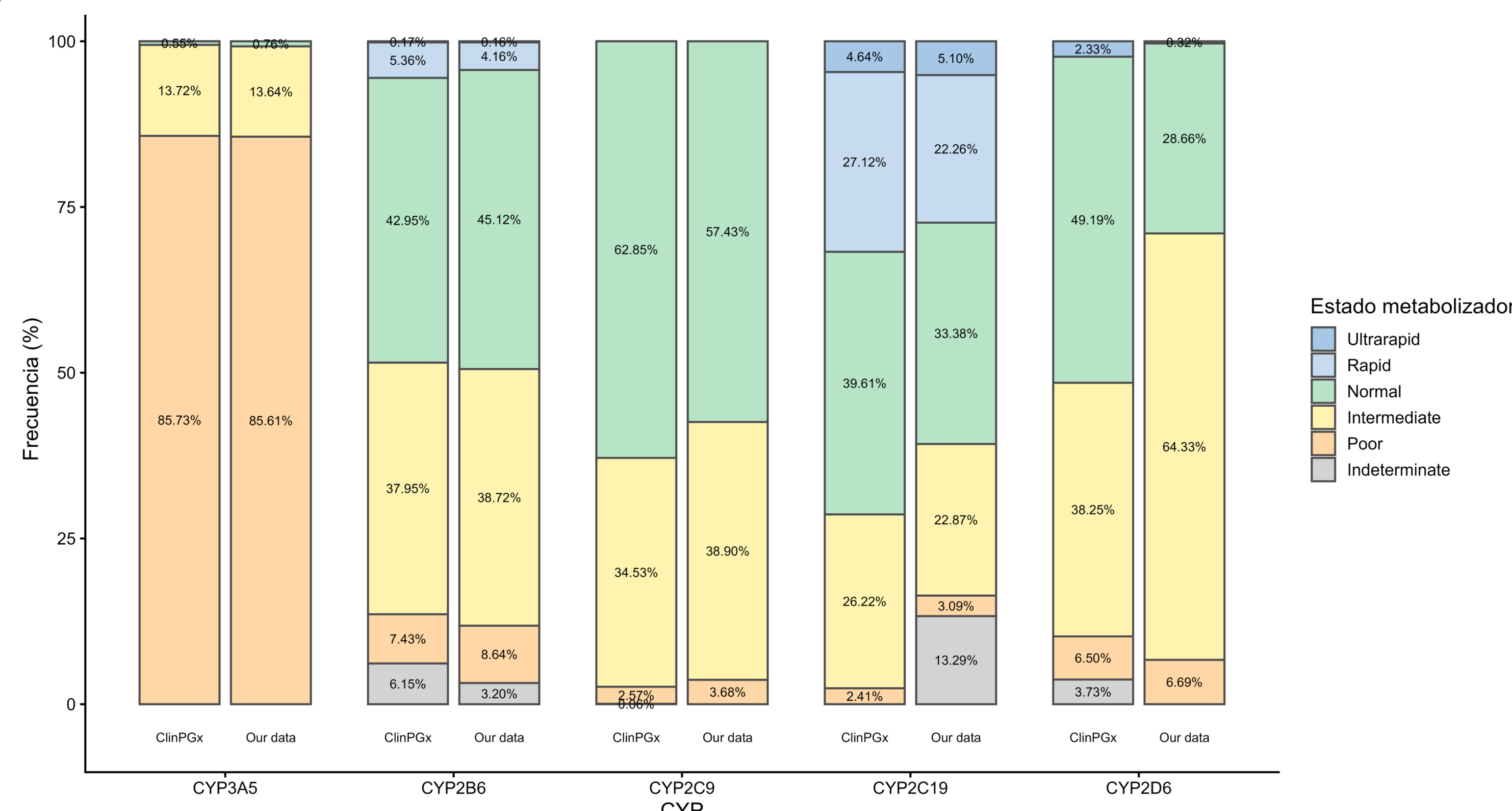


Figure 3. Comparison of CYP enzyme metabolizer phenotype frequencies between our cohort (derived using the GenoStaR R package) and ClinPGx population data. Stacked bars show the proportion (%) of ultrarapid, rapid, normal, intermediate, poor, and indeterminate metabolizers for each CYP enzyme.

Discussion

GenoStaR represents a fast and useful tool for pharmacogenetic prediction of metabolic phenotypes from genotyping data, although further improvements are required to increase call rate and accuracy for *CYP2D6*.

References

- Coulter, A.J et al., (2026) GenoStaR: An R Package for Genotype to Star Allele Conversion for Major Cytochrome P450 Family of Genes. *Clinical Pharmacology & Therapeutics*, 119(1), 74–79.
- ClinPGx. Pharmacogenomics Allele Definition Tables and Gene Reference Resources. Available at: <https://www.clinpgx.org>
- gnomAD. Genome Aggregation Database. Used to retrieve population allele frequencies for the variants analyzed. Available at: <https://gnomad.broadinstitute.org>